Week 7 In-Class Exercise

Adverse events, explanations and the threshold

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Monday, October 26, 2026 · 25 minutes · pairs

Part A — Eight notes by hand (10 minutes)

For each call note decide: (1) is an adverse event mentioned? (2) are the four elements present — identifiable patient, identifiable reporter, suspect drug, event? (3) serious? (4) expected (in label)? (5) what is the reporting path and, if serious and unexpected, the due date? Write the one phrase that decided it. Use the lexicon: in-label events include nausea, diarrhoea, hypoglycaemia, headache, dizziness, UTI, rash; pancreatitis, kidney injury, angioedema and syncope are not in label.

Note 1. HCP has had no reports of dizziness in pts on GLUCOREN. Tolerability good overall. Left samples.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 2. One of her patients, 63 F w/ CKD3, developed loose stools within 2 wks of starting. Dose held. HCP will reassess.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 3. HCP asked how common pancreatitis is with GLUCOREN after reading about it with another drug in the class. No patient discussed. Referred to MI.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 4. Serious case per HCP: a 71-year-old male admitted last week with pancreatitis three weeks after starting GLUCOREN. Drug stopped. HCP asked me to report it.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 5. Pt starting GLUCOREN next week is nervous about nausea after reading the forum. HCP counselled; no issues yet.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 6. BCBS moved GLUCOREN to tier 3. Two pts switched to CARDIVEX. HCP says one of them had been complaining of headaches on GLUCOREN anyway.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 7. Mrs. Alvarez (DOB 03/14/1961) called the office: sugar in the 50s twice since the dose increase. HCP reduced the dose.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Note 8. HCP mentioned a CARDIVEX patient who fainted last month. Not our product. Discussed GLUCOREN's CV data on-label.

AE?	Four elements?	Serious?	In label?	Path / due date	Deciding phrase

Part B — Read an explanation (8 minutes)

Two notes were flagged by the detector. For each you see the note and its top word contributions. Decide: would you trust the flag? would you send the model back — and why? what behavioural probe would you add to the retrain checklist?

Explanation 1 — flagged, p(AE) = 0.91

"Dr. reports CVS Caremark is requiring prior authorization; two patients waiting over a week. Also mentioned one of his patients dropped off GLUCOREN due to stomach upset."

Top contributions: "caremark is" +1.2 · "one of his" +0.9 · "dropped off" +0.8 · "stomach upset" +0.6 · "prior authorization" −0.4

Trust? ☐ yes ☐ no Send back? ☐ yes ☐ no Why / probe:

_____

Explanation 2 — flagged, p(AE) = 0.62

"HCP wants patient education materials on hypoglycemia awareness for patients also on insulin. Also reviewed the copay program."

Top contributions: "hypoglycemia" +0.9 · "patients" +0.5 · "insulin" +0.3 · "education materials" −0.5 · "wants" −0.3

Trust? ☐ yes ☐ no Send back? ☐ yes ☐ no Why / probe:

_____

Part C — Set the threshold (7 minutes)

The cost table from tonight's notebook (about 84 notes a week; six reviewer-minutes per note):

Threshold	Recall	Precision	Flags / week	Reviewer hours / week	AEs missed / year
0.20	97%	62%	13	1.3	12
0.30	97%	80%	10	1.0	14
0.47	95%	95%	8	0.8	20
0.50	93%	94%	8	0.8	29
0.70	79%	95%	7	0.7	89

Safety has given you a reviewer budget of 2 hours a week. (1) Which threshold do you pick, and how many AEs a year does it miss? (2) Next year GLUCOREN launches a second indication and the field grows: three times the notes, same mix. Which threshold now, and what do you ask Safety for? (3) Write the one sentence that justifies your choice to the head of PV.

(1) _____

(2) _____

(3) _____